

First-Principles Derivations and Calculations

V-Dipole Antenna Analysis for Meteor M2-4 LRPT

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1 Fundamental Physical Constants

All calculations utilize exact CODATA physical constants to eliminate truncation errors:

- Speed of light in vacuum: $c = 299,792,458$ m/s
- Vacuum permittivity: $\varepsilon_0 = 8.8541878128 \times 10^{-12}$ F/m
- Earth mean radius: $R_E = 6,371$ km
- Meteor M2-4 altitude: $h_s = 826.91$ km
- LRPT Downlink frequency: $f_0 = 137.9 \times 10^6$ Hz

2 Antenna Geometry and Dimensions

2.1 Equation 1: Free-Space Wavelength

$$\lambda = \frac{c}{f_0} = \frac{299,792,458}{137,900,000} \quad (1)$$

$$\lambda = \mathbf{2.17398} \text{ m} \quad (2)$$

2.2 Equation 2: Quarter-Wave Arm Length

$$\ell_0 = \frac{\lambda}{4} = \frac{2.17398}{4} \quad (3)$$

$$\ell_0 = \mathbf{0.54350} \text{ m} \quad (4)$$

2.3 Equation 3: Physical Shortening Correction

Applying the empirically validated community standard arm length of 53.4 cm:

$$\ell = k_s \cdot \ell_0 \implies 0.53400 = k_s \cdot 0.54350 \quad (5)$$

$$k_s = \mathbf{0.98252} \quad (6)$$

$$\ell = \mathbf{0.53400} \text{ m} \quad (7)$$

3 Orbital Mechanics and Slant Range

3.1 Equation 4: Exact Spherical Slant Range

Using the exact geometric line-of-sight formula $d(\varepsilon) = \sqrt{(R_E + h_s)^2 - R_E^2 \cos^2 \varepsilon} - R_E \sin \varepsilon$:

- **At $\varepsilon = 90^\circ$ (Zenith):** $d = \sqrt{(7197.91)^2 - 0} - 6371 = \mathbf{826.91000}$ km
- **At $\varepsilon = 30^\circ$:** $d = \sqrt{51809908.57 - (6371 \times 0.866025)^2} - (6371 \times 0.5) = \mathbf{1437.01856}$ km
- **At $\varepsilon = 10^\circ$:** $d = \sqrt{51809908.57 - (6371 \times 0.984807)^2} - (6371 \times 0.173648) = \mathbf{2421.32053}$ km
- **At $\varepsilon = 5^\circ$:** $d = \sqrt{51809908.57 - (6371 \times 0.996194)^2} - (6371 \times 0.087155) = \mathbf{2840.10876}$ km

4 Free-Space Path Loss (FSPL)

4.1 Equation 5: Analytic FSPL

The exact FSPL formula is $\text{FSPL} = 20 \log_{10}(d) + 20 \log_{10}(f_0) + 20 \log_{10}\left(\frac{4\pi}{c}\right)$. First, we derive the exact unit-conversion constant:

$$20 \log_{10}\left(\frac{4\pi}{299792458}\right) = 20 \log_{10}(4.18879 \times 10^{-8}) = -\mathbf{147.55222} \text{ dB} \quad (8)$$

Now, applying this to the slant ranges (with d in meters):

- **90°:** $20 \log_{10}(826910) + 162.79844 - 147.55222 = \mathbf{133.59210} \text{ dB}$
- **30°:** $20 \log_{10}(1437018.56) + 162.79844 - 147.55222 = \mathbf{138.39477} \text{ dB}$
- **10°:** $20 \log_{10}(2421320.53) + 162.79844 - 147.55222 = \mathbf{142.92576} \text{ dB}$
- **5°:** $20 \log_{10}(2840108.76) + 162.79844 - 147.55222 = \mathbf{144.31306} \text{ dB}$

5 Ground Reflection and Fresnel Coefficients

5.1 Equation 6: Complex Permittivity

For Lahore alluvial soil ($\varepsilon_r = 13$, $\sigma = 0.006 \text{ S/m}$):

$$\tilde{\varepsilon}_c = \varepsilon_r - j \frac{\sigma}{2\pi f_0 \varepsilon_0} \quad (9)$$

$$\tilde{\varepsilon}_c = 13 - j \frac{0.006}{2\pi(137.9 \times 10^6)(8.8541878 \times 10^{-12})} \quad (10)$$

$$\tilde{\varepsilon}_c = 13 - j \frac{0.006}{0.0076716} = \mathbf{13 - j0.78209} \quad (11)$$

5.2 Equation 7: Horizontal Fresnel Reflection Coefficient

Evaluating $\Gamma_h(\varepsilon) = \frac{\sin \varepsilon - \sqrt{\tilde{\varepsilon}_c - \cos^2 \varepsilon}}{\sin \varepsilon + \sqrt{\tilde{\varepsilon}_c - \cos^2 \varepsilon}}$ using exact complex arithmetic:

- **At 5°:** $|\Gamma_h| = \mathbf{0.95101}$, $\angle \Gamma_h = \mathbf{179.90633}^\circ$
- **At 30°:** $|\Gamma_h| = \mathbf{0.75034}$, $\angle \Gamma_h = \mathbf{179.46793}^\circ$
- **At 90° (Zenith):** $|\Gamma_h| = \mathbf{0.56623}$, $\angle \Gamma_h = \mathbf{178.96692}^\circ$
(Note: The reflection at zenith is not zero; it is governed by the intrinsic impedance mismatch, yielding ≈ -0.566 . The zenith null is caused by destructive phase interference, not zero reflection.)

5.3 Equation 8: Ground Array Factor

Evaluating $|1 + \Gamma_h e^{j2kh \sin \varepsilon}|^2$ at optimal height $h = \lambda/2 = 1.08699 \text{ m}$:

- **At 5°:** **0.27892** (-5.54523 dB) — *Destructive near horizon*
- **At 30°:** **3.06362** ($+4.86235 \text{ dB}$) — *Constructive peak*
- **At 90°:** **0.18834** (-7.25051 dB) — *Destructive at zenith*

6 Radiation Pattern Functions

6.1 Equation 9 & 10: Hertzian and Half-Wave Dipole

The half-wave dipole element factor $F(\theta) = \frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta}$ evaluates to:

- At 90° (Broadside): **1.00000**
- At 45° : **0.62786**
- At 0° (End-fire): **0.00000** (Limit)

6.2 Equation 11: V-Dipole Azimuth Pattern

The unnormalized azimuth function $G(\phi) = \sin^2 \phi + \sin^2(30^\circ) \cos^2 \phi$ evaluates to:

- At 90° (Broadside): $1 + (0.25 \times 0) = \mathbf{1.00000}$
- At 45° : $0.5 + (0.25 \times 0.5) = \mathbf{0.62500}$
- At 0° (End-fire): $0 + (0.25 \times 1) = \mathbf{0.25000}$

6.3 Equation 12: Peak Elevation Angle

Using the closed-form approximation $\varepsilon_{\max} = \arcsin\left(\frac{\lambda}{4h}\right)$:

- $h = 1.08699 \text{ m } (\lambda/2)$: $\arcsin(0.50000) = \mathbf{30.00000^\circ}$
- $h = 1.50000 \text{ m}$: $\arcsin(0.36233) = \mathbf{21.24385^\circ}$
- $h = 2.50000 \text{ m}$: $\arcsin(0.21740) = \mathbf{12.55985^\circ}$

7 Impedance, Polarisation, and Doppler

7.1 Equation 13: VSWR

Using simulated feed impedance $Z_{\text{in}} = 56.2 \Omega$:

$$\Gamma = \frac{56.2 - 50}{56.2 + 50} = \frac{6.2}{106.2} = \mathbf{0.05838} \quad (12)$$

$$\text{VSWR} = \frac{1 + 0.05838}{1 - 0.05838} = \mathbf{1.12399} \quad (13)$$

7.2 Equation 14: Polarisation Loss Factor

Exact loss for RHCP satellite transmission received by a linearly polarized V-Dipole:

$$L_{\text{pol}} = \left| \frac{1}{\sqrt{2}} \right|^2 = \mathbf{0.50000} \implies 10 \log_{10}(0.5) = \mathbf{-3.01030 \text{ dB}} \quad (14)$$

7.3 Equation 15: Doppler Shift

Using maximum radial velocity $v_r = 7420 \text{ m/s}$:

$$\Delta f_D = \frac{7420}{299792458} \times 137.9 \times 10^6 = 3412.84519 \text{ Hz} = \mathbf{+3.41285 \text{ kHz}} \quad (15)$$

8 System Noise and Atmospheric Attenuation

8.1 Equation 16: Cascaded Noise Figure (Friis Formula)

Converting to exact linear scale ($F = 10^{\text{NF}/10}$, $G = 10^{\text{Gain}/10}$):

- **Stage 1 (LNA):** $F_1 = 1.17490$, $G_1 = 100.00000$
- **Stage 2 (Cable):** $F_2 = 1.27353$, $G_2 = 0.78524$
- **Stage 3 (SDR):** $F_3 = 2.23872$

With LNA first:

$$F_{\text{sys}} = 1.17490 + \frac{0.27353}{100} + \frac{1.23872}{78.524} = 1.19342 \implies \mathbf{0.76798 \text{ dB}} \quad (16)$$

Without LNA (Cable first):

$$F_{\text{sys}} = 1.27353 + \frac{1.23872}{0.78524} = 2.85103 \implies \mathbf{4.54999 \text{ dB}} \quad (17)$$

8.2 Equation 17: Atmospheric Attenuation Scaling

Using zenith loss $L_{\text{atm},90^\circ} = 0.15 \text{ dB}$:

$$L_{\text{atm}}(10^\circ) = \frac{0.15}{\sin(10^\circ)} = \frac{0.15}{0.173648} = \mathbf{0.86382 \text{ dB}} \quad (18)$$

9 Complete End-to-End Link Budget

Utilizing the exact first-principles derivations at $\varepsilon = 30^\circ$:

Parameter	Exact Value	Unit
Satellite TX EIRP (estimated)	+4.00000	dBW
Free-Space Path Loss (30°)	-138.39477	dB
Atmospheric + ionospheric loss	-0.40000	dB
V-Dipole antenna gain (30°)	+2.20000	dB
Polarisation mismatch loss	-3.01030	dB
Cable loss (3m RG-58)	-1.05000	dB
Signal at LNA input	-136.65507	dBW
LNA Gain	+20.00000	dB
Signal at SDR input	-116.65507	dBW (-86.65 dBm)
System noise floor (150 kHz BW)	-121.44000	dBm
SNR at SDR input	+34.78493	dB
Required E_b/N_0 for OQPSK	≈ 10.60000	dB
Link margin	+24.18493	dB (Link Closes)

Conclusion of Derivations

The exact analytical calculations prove the true FSPL at 30° is **138.39 dB**. Consequently, the signal arrives at the SDR at a highly robust **−86.65 dBm**, yielding a massive **+24.18 dB** link margin that easily clears the RTL-SDR noise floor and guarantees successful LRPT image decoding.